

Dmitry Kovalev

PERSONAL DATA

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POSITIONS

2023-Now	Senior Researcher Yandex Research , Moscow, Russia
2021-2023	Research Intern Research Center for Trusted Artificial Intelligence Ivannikov Institute for System Programming , Moscow, Russia
2022-2023	Postdoctoral Researcher Université catholique de Louvain , Louvain-la-Neuve, Belgium
2022	Researcher Laboratory of Advanced Combinatorics and Network Applications Moscow Institute of Physics and Technology , Dolgoprudny, Russia

EDUCATION

2019-2022	PhD in Computer Science King Abdullah University of Science and Technology , Thuwal, Saudi Arabia Advisor: Peter Richtárik Thesis: “ Optimal Algorithms for Affinely Constrained, Distributed, Decentralized, Minimax, and High-Order Optimization Problems ” Defense Committee: – External: Yurii Nesterov , Arkadi Nemirovsky – Internal: David E. Keyes , Di Wang , Matteo Parsani
2018-2021	MS in Applied Mathematics and Physics Moscow Institute of Physics and Technology , Dolgoprudny, Russia Advisor: Alexander Gasnikov
2018-2019	MS in Computer Science King Abdullah University of Science and Technology , Thuwal, Saudi Arabia Advisor: Peter Richtárik
2014-2018	BS in Applied Mathematics and Physics Moscow Institute of Physics and Technology , Dolgoprudny, Russia Advisor: Alexander Gasnikov

RESEARCH INTERESTS

Optimization, Federated and Distributed Learning
Machine Learning, Deep Learning

SKILLS

PROGRAMMING	C/C++, Python, Algorithms and Data Structures (Codeforces master), JAX, PyTorch; PAST EXPERIENCE: Go, C#, VB.NET, SQL, Julia, DirectX, Vulkan, HTML/CSS
MATHEMATICS	Calculus, Linear Algebra, Probability and Statistics, Convex Analysis
COMPUTER	macOS, LaTeX, Git
LANGUAGES	English (Advanced), Russian (Native)

HONORS AND AWARDS

1. **Charles Broyden Prize 2023**, Optimization Methods and Software Editorial Board, 2026
2. **National Award “AI Leaders” 2025**, AI Journey 2025 Conference
3. **Boris Polyak Scientific Prize for Young Researchers 2025**, Innopolis University
4. **CEMSE Student Research Excellence Award**, King Abdullah University of Science and Technology, 2021
5. **Best Student Paper Award at FL-ICML 2021 Workshop**
6. **Ilya Segalovich Scientific Prize for Young Researchers 2021**, Yandex
7. **Annual PhD progress marked as Outstanding**, King Abdullah University of Science and Technology, 2019-2021
8. **Ilya Segalovich Scientific Prize for Young Researchers 2020**, Yandex
9. **Dean’s Award**, given to top students accepted to King Abdullah University of Science and Technology, 2018
10. **Abramov’s Fund Scholarship for Excellence in Study**, Moscow Institute of Physics and Technology, 2015-2017
11. **Asian Physics Olympiad (APhO) 2014, Honourable Mention**, Singapore (participated as a member of Russian national team)
12. **Russian President’s Scholarship for High School Students**, given for prize-winning at final round of All-Russian School Olympiads, 2012-2014
13. **Moscow Governor’s Scholarship for High School Students**, given for prize-winning at region and final rounds of All-Russian School Olympiads, 2012-2014
14. **All-Russian School Physics Olympiad, Final Round Prize-Winner**, Saint-Petersburg, 2014
15. **All-Russian School Programming Olympiad, Region Round Winner**, Moscow, 2014
16. **All-Russian School Math Olympiad, Region Round Winner**, Moscow, 2014
17. **All-Russian School Physics Olympiad, Final Round Winner**, Vladivostok, 2013
18. **All-Russian School Physics Olympiad, Final Round Prize-Winner**, Saransk, 2012

PUBLICATIONS

1. **Decentralized Optimization with Coupled Constraints** (Demyan Yarmoshik, Alexander Rogozin, Nikita Kiselev, Daniil Dorin, Alexander Gasnikov, Dmitry Kovalev). *International Conference on Learning Representations*, 2025.
2. **On Linear Convergence in Smooth Convex-Concave Bilinearly-Coupled Saddle-Point Optimization: Lower Bounds and Optimal Algorithms** (Ekaterina Borodich, Alexander Gasnikov, Dmitry Kovalev). *International Conference on Machine Learning*, 2025.
3. **Decentralized saddle point problems via non-Euclidean mirror prox** (Alexander Rogozin, Aleksandr Beznosikov, Darina Dvinskikh, Dmitry Kovalev, Pavel Dvurechensky, Alexander Gasnikov). *Optimization Methods and Software*, 2025.
4. **Near-optimal algorithm with complexity separation for strongly convex-strongly concave composite saddle point problems** (Ekaterina Borodich, Georgiy Kormakov, Dmitry Kovalev, Aleksandr Beznosikov, Alexander Gasnikov). *Optimization Methods and Software*, 2025.
5. **An Optimal Algorithm for Strongly Convex Min-Min Optimization** (Dmitry Kovalev, Alexander Gasnikov, Grigory Malinovsky). *Uncertainty in Artificial Intelligence*, 2025.

6. **Lower Bounds and Optimal Algorithms for Non-Smooth Convex Decentralized Optimization over Time-Varying Networks** (Dmitry Kovalev, Ekaterina Borodich, Alexander Gasnikov, Dmitrii Feoktistov). *Advances in Neural Information Processing Systems*, 2024.
7. **Decentralized convex optimization on time-varying networks with application to Wasserstein barycenters** (Olga Yufereva, Michael Persiianov, Pavel Dvurechensky, Alexander Gasnikov, Dmitry Kovalev). *Computational Management Science*, 2024.
8. **Decentralized saddle-point problems with different constants of strong convexity and strong concavity** (Dmitry Metelev, Alexander Rogozin, Alexander Gasnikov, Dmitry Kovalev). *Computational Management Science*, 2024.
9. **Convex-Concave Interpolation and Application of PEP to the Bilinear-Coupled Saddle Point Problem** (Valery Krivchenko, Alexander Gasnikov, Dmitry Kovalev). *Russian Journal of Nonlinear Dynamics*, 2024.
10. **Non-smooth setting of stochastic decentralized convex optimization problem over time-varying graphs** (Aleksandr Lobanov, Andrew Veprikov, Georgiy Konin, Aleksandr Beznosikov, Alexander Gasnikov, Dmitry Kovalev). *Computational Management Science*, 2023.
11. **Decentralized convex optimization over time-varying graphs** (Alexander Rogozin, Alexander Gasnikov, Aleksandr Beznosikov, Dmitry Kovalev). *Encyclopedia of Optimization*, 2023.
12. **Smooth monotone stochastic variational inequalities and saddle point problems: A survey** (Aleksandr Beznosikov, Boris Polyak, Eduard Gorbunov, Dmitry Kovalev, Alexander Gasnikov). *European Mathematical Society Magazine*, 2023.
13. **Is consensus acceleration possible in decentralized optimization over slowly time-varying networks?** (Dmitry Metelev, Alexander Rogozin, Dmitry Kovalev, Alexander Gasnikov). *International Conference on Machine Learning*, 2023.
14. **Stochastic distributed learning with gradient quantization and double-variance reduction** (Samuel Horvath, Dmitry Kovalev, Konstantin Mishchenko, Peter Richtarik, Sebastian Stich). *Optimization Methods and Software*, 2023.
15. **Accelerated primal-dual gradient method for smooth and convex-concave saddle-point problems with bilinear coupling** (Dmitry Kovalev, Alexander Gasnikov, Peter Richtarik). *Advances in Neural Information Processing Systems*, 2022.
16. **Communication acceleration of local gradient methods via an accelerated primal-dual algorithm with an inexact prox** (Abdurakhmon Sadiev, Dmitry Kovalev, Peter Richtarik). *Advances in Neural Information Processing Systems*, 2022.
17. **Optimal algorithms for decentralized stochastic variational inequalities** (Dmitry Kovalev, Aleksandr Beznosikov, Abdurakhmon Sadiev, Michael Persiianov, Peter Richtarik, Alexander Gasnikov). *Advances in Neural Information Processing Systems*, 2022.
18. **Optimal gradient sliding and its application to optimal distributed optimization under similarity** (Dmitry Kovalev, Aleksandr Beznosikov, Ekaterina Borodich, Alexander Gasnikov, Gesualdo Scutari). *Advances in Neural Information Processing Systems*, 2022.
19. **The first optimal acceleration of high-order methods in smooth convex optimization** (Dmitry Kovalev, Alexander Gasnikov). *Advances in Neural Information Processing Systems*, 2022.
20. **The first optimal algorithm for smooth and strongly-convex-strongly-concave minimax optimization** (Dmitry Kovalev, Alexander Gasnikov). *Advances in Neural Information Processing Systems*, 2022.

21. **Accelerated variance-reduced methods for saddle-point problems** (Ekaterina Borodich, Vladislav Tominin, Yaroslav Tominin, Dmitry Kovalev, Alexander Gasnikov, Pavel Dvurechensky). *EURO Journal on Computational Optimization*, 2022.
22. **An optimal algorithm for strongly convex minimization under affine constraints** (Adil Salim, Laurent Condat, Dmitry Kovalev, Peter Richtarik). *International Conference on Artificial Intelligence and Statistics*, 2022.
23. **IntSGD: Adaptive floatless compression of stochastic gradients** (Konstantin Mishchenko, Bokun Wang, Dmitry Kovalev, Peter Richtarik). *International Conference on Learning Representations*, 2022.
24. **Lower bounds and optimal algorithms for smooth and strongly convex decentralized optimization over time-varying networks** (Dmitry Kovalev, Elnur Gasanov, Alexander Gasnikov, Peter Richtarik). *Advances in Neural Information Processing Systems*, 2021.
25. **A linearly convergent algorithm for decentralized optimization: Sending less bits for free!** (Dmitry Kovalev, Anastasia Koloskova, Martin Jaggi, Peter Richtarik, Sebastian Stich). *International Conference on Artificial Intelligence and Statistics*, 2021.
26. **ADOM: accelerated decentralized optimization method for time-varying networks** (Dmitry Kovalev, Egor Shulgin, Peter Richtarik, Alexander V Rogozin, Alexander Gasnikov). *International Conference on Machine Learning*, 2021.
27. **Near-optimal decentralized algorithms for saddle point problems over time-varying networks** (Aleksandr Beznosikov, Alexander Rogozin, Dmitry Kovalev, Alexander Gasnikov). *International Conference on Optimization and Applications*, 2021.
28. **Towards accelerated rates for distributed optimization over time-varying networks** (Alexander Rogozin, Vladislav Lukoshkin, Alexander Gasnikov, Dmitry Kovalev, Egor Shulgin). *International Conference on Optimization and Applications*, 2021.
29. **Linearly converging error compensated SGD** (Eduard Gorbunov, Dmitry Kovalev, Dmitry Makarenko, Peter Richtarik). *Advances in Neural Information Processing Systems*, 2020.
30. **Optimal and practical algorithms for smooth and strongly convex decentralized optimization** (Dmitry Kovalev, Adil Salim, Peter Richtarik). *Advances in Neural Information Processing Systems*, 2020.
31. **Don't jump through hoops and remove those loops: SVRG and Katyusha are better without the outer loop** (Dmitry Kovalev, Samuel Horvath, Peter Richtarik). *Algorithmic Learning Theory*, 2020.
32. **Accelerated methods for saddle-point problem** (Mohammad S Alkousa, Alexander Vladimirovich Gasnikov, Darina Mikhailovna Dvinskikh, Dmitry A Kovalev, Fedor Sergeevich Stonyakin). *Computational Mathematics and Mathematical Physics*, 2020.
33. **Revisiting stochastic extragradient** (Konstantin Mishchenko, Dmitry Kovalev, Egor Shulgin, Peter Richtarik, Yura Malitsky). *International Conference on Artificial Intelligence and Statistics*, 2020.
34. **Acceleration for compressed gradient descent in distributed and federated optimization** (Zhize Li, Dmitry Kovalev, Xun Qian, Peter Richtarik). *International Conference on Machine Learning*, 2020.
35. **From local SGD to local fixed-point methods for federated learning** (Grigory Malinovskiy, Dmitry Kovalev, Elnur Gasanov, Laurent Condat, Peter Richtarik). *International Conference on Machine Learning*, 2020.
36. **Variance reduced coordinate descent with acceleration: New method with a surprising application to finite-sum problems** (Filip Hanzely, Dmitry Kovalev, Peter Richtarik). *International Conference on Machine Learning*, 2020.

37. **RSN: randomized subspace Newton** (Robert Gower, Dmitry Kovalev, Felix Lieder, Peter Richtarik). *Advances in Neural Information Processing Systems*, 2019.
38. **Stochastic proximal Langevin algorithm: Potential splitting and nonasymptotic rates** (Adil Salim, Dmitry Kovalev, Peter Richtarik). *Advances in Neural Information Processing Systems*, 2019.
39. **Stochastic spectral and conjugate descent methods** (Dmitry Kovalev, Peter Richtarik, Eduard Gorbunov, Elnur Gasanov). *Advances in Neural Information Processing Systems*, 2018.
40. **A hypothesis about the rate of global convergence for optimal methods (Newton's type) in smooth convex optimization** (Alexander Gasnikov, Dmitry Kovalev). *Computer research and modeling*, 2018.

PREPRINTS

1. **Optimal Projection-Free Adaptive SGD for Matrix Optimization** (Dmitry Kovalev). *arXiv preprint arXiv:2604.02505*, 2026.
2. **Strengthening the finite characterizations of smooth min-max games** (Valery Krivchenko, Alexander Gasnikov, Dmitry Kovalev). *arXiv preprint arXiv:2603.17053*, 2026.
3. **Muon is Provably Faster with Momentum Variance Reduction** (Xun Qian, Hussein Rammal, Dmitry Kovalev, Peter Richtarik). *arXiv preprint arXiv:2512.16598*, 2025.
4. **Non-Euclidean SGD for Structured Optimization: Unified Analysis and Improved Rates** (Dmitry Kovalev, Ekaterina Borodich). *arXiv preprint arXiv:2511.11466*, 2025.
5. **Nesterov finds GRAAL: Optimal and adaptive gradient method for convex optimization** (Ekaterina Borodich, Dmitry Kovalev). *arXiv preprint arXiv:2507.09823*, 2025.
6. **Sgd with adaptive preconditioning: Unified analysis and momentum acceleration** (Dmitry Kovalev). *arXiv preprint arXiv:2506.23803*, 2025.
7. **Understanding gradient orthogonalization for deep learning via non-euclidean trust-region optimization** (Dmitry Kovalev). *arXiv preprint arXiv:2503.12645*, 2025.
8. **On solving minimization and min-max problems by first-order methods with relative error in gradients** (Artem Vasin, Valery Krivchenko, Dmitry Kovalev, Fedor Stonyakin, Nazarii Tupitsa, Pavel Dvurechensky, Mohammad Alkousa, Nikita Kornilov, Alexander Gasnikov). *arXiv preprint arXiv:2503.06628*, 2025.
9. **Decentralized finite-sum optimization over time-varying networks** (Dmitry Metelev, Savelii Chezhegov, Alexander Rogozin, Aleksandr Beznosikov, Alexander Sholokhov, Alexander Gasnikov, Dmitry Kovalev). *arXiv preprint arXiv:2402.02490*, 2024.
10. **On scaled methods for saddle point problems** (Aleksandr Beznosikov, Aibek Alanov, Dmitry Kovalev, Martin Takac, Alexander Gasnikov). *arXiv preprint arXiv:2206.08303*, 2022.
11. **Decentralized distributed optimization for saddle point problems** (Alexander Rogozin, Aleksandr Beznosikov, Darina Dvinskikh, Dmitry Kovalev, Pavel Dvurechensky, Alexander Gasnikov). *arXiv preprint arXiv:2102.07758*, 2021.
12. **Fast linear convergence of randomized BFGS** (Dmitry Kovalev, Robert M Gower, Peter Richtarik, Alexander Rogozin). *arXiv preprint arXiv:2002.11337*, 2020.
13. **Distributed fixed point methods with compressed iterates** (Selim Chraibi, Ahmed Khaled, Dmitry Kovalev, Peter Richtarik, Adil Salim, Martin Takac). *arXiv preprint arXiv:1912.09925*, 2019.

14. **Stochastic Newton and cubic Newton methods with simple local linear-quadratic rates** (Dmitry Kovalev, Konstantin Mishchenko, Peter Richtarik). *arXiv preprint arXiv:1912.01597*, 2019.

CONFERENCE POSTERS AND TALKS

1. **Talk: Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization**, *HDI Lab Structural Learning Seminar*, Higher School of Economics, Moscow, Russia (December 2025)
2. **Talk: Recent Advances in Adaptive Convex Optimization**, *Mathematical Foundations of Artificial Intelligence Seminar*, Steklov Mathematical Institute, Moscow, Russia (November 2025)
3. **Talk: Recent Advances in Adaptive Convex Optimization**, *International Conference on Computational Optimization 2025*, Abu Dhabi, United Arab Emirates (October 2025)
4. **Talk: Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization**, *Traditional School (Control, Information and Optimization)*, Innopolis University, Kazan, Russia (June 2025)
5. **Talk: Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization**, *Data Fusion 2025 Conference*, Moscow, Russia (April 2025)
6. **Talk: Lower Bounds and Optimal Algorithms for Smooth and Strongly Convex Decentralized Optimization Over Time-Varying Networks**, *Rising Stars in AI Symposium 2022*, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia (March 2022)
7. **Talk/Poster: Lower Bounds and Optimal Algorithms for Smooth and Strongly Convex Decentralized Optimization Over Time-Varying Networks**, *NeurIPS 2021*, Online (December 2021)
8. **Talk/Poster: Lower Bounds and Optimal Algorithms for Smooth and Strongly Convex Decentralized Optimization Over Time-Varying Networks**, *International Workshop on Federated Learning for User Privacy and Data Confidentiality in Conjunction with ICML 2021*, Online (July 2021)
9. **Talk/Poster: ADOM: Accelerated Decentralized Optimization Method for Time-Varying Networks**, *ICML 2021*, Online (July 2021)
10. **Poster: ADOM: Accelerated Decentralized Optimization Method for Time-Varying Networks**, *Optimization Without Borders Conference*, Sirius University, Sochi, Russia (July 2021)
11. **Talk/Poster: A Linearly Convergent Algorithm for Decentralized Optimization: Sending Less Bits for Free!**, *AISTATS 2021*, Online (April 2021)
12. **Poster: Linearly Converging Error Compensated SGD**, *NeurIPS 2020*, Online (December 2020)
13. **Talk/Poster: Optimal and Practical Algorithms for Smooth and Strongly Convex Decentralized Optimization**, *NeurIPS 2020*, Online (December 2020)
14. **Talk: Variance Reduced Coordinate Descent with Acceleration: New Method With a Surprising Application to Finite-Sum Problems**, *ICML 2020*, Online (July 2020)
15. **Poster: RSN: Randomized Subspace Newton**, *NeurIPS 2019*, Vancouver, Canada (December 2019)
16. **Poster: Stochastic Proximal Langevin Algorithm: Potential Splitting and Nonasymptotic Rates**, *NeurIPS 2019*, Vancouver, Canada (December 2019)
17. **Talk: Revisiting Stochastic Extragradient Method**, *International Conference on Continuous Optimization 2019*, Technical University, Berlin, Germany (August 2019)

18. **Poster: Stochastic Distributed Learning with Gradient Quantization and Variance Reduction**, *Data Science Summer School 2019*, Ecole Polytechnique, Paris, France (June 2019)
19. **Poster: Stochastic Distributed Learning with Gradient Quantization and Variance Reduction**, *Traditional School (Control, Information and Optimization)*, Higher School of Economics Study Center, Voronovo, Russia (June 2019)
20. **Talk: Stochastic Spectral Descent Methods**, *Automatic control and Optimization Theory Weekly Seminar*, Institute for Control Problems, Moscow, Russia (March 2019)
21. **Talk: Stochastic Distributed Learning with Gradient Quantization and Variance Reduction**, *Modern Optimization Methods Seminar*, Moscow Institute of Physics and Technology, Moscow, Russia (March 2019)
22. **Poster: Stochastic Spectral Descent Methods**, *NeurIPS 2018*, Montreal, Canada (December 2018)
23. **Poster: Stochastic Spectral Descent Methods**, *Optimization and Big Data Workshop*, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia (June 2018)
24. **Poster: Stochastic Spectral Descent Methods**, *Traditional School (Control, Information and Optimization)*, Higher School of Economics Study Center, Voronovo, Russia (February 2018)

TEACHING EXPERIENCE

1. **Mentor for a Research Project with MIPT student Ekaterina Borodich** (online, led to NeurIPS 2022 paper “Optimal Gradient Sliding and its Application to Distributed Optimization Under Similarity”), Moscow Institute of Physics and Technology, Moscow, Russia (March 2022)
2. **Mentor for a Research Project with KAUST Student Abdurakhmon Sadiev** (led to NeurIPS 2022 paper “Communication Acceleration of Local Gradient Methods via an Accelerated Primal-Dual Algorithm with Inexact Prox”), King Abdullah University of Science and Technology, Thuwal, Saudi Arabia (March 2022)
3. **Project Mentor at «Modern Information, Optimization and Control Methods» Student Educational Program**, Sirius University, Sochi, Russia (July-August 2021)
4. **Mentor for a Research Project with Student Grigory Malinovsky** (done during his internship at KAUST, led to ICML 2020 paper “From Local SGD to Local Fixed Point Methods for Federated Learning”), King Abdullah University of Science and Technology, Thuwal, Saudi Arabia (January 2020)

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